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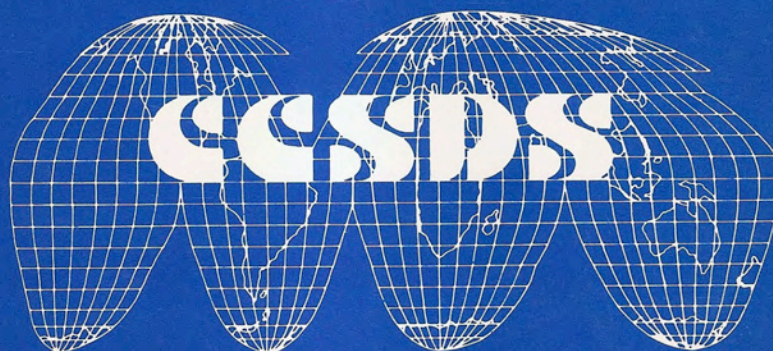
**CONSULTATIVE
COMMITTEE FOR
SPACE DATA SYSTEMS**

RECOMMENDATION FOR SPACE
DATA SYSTEM STANDARDS:

**TELEMETRY
CHANNEL
CODING**

"BLUE BOOK"

MAY 1984



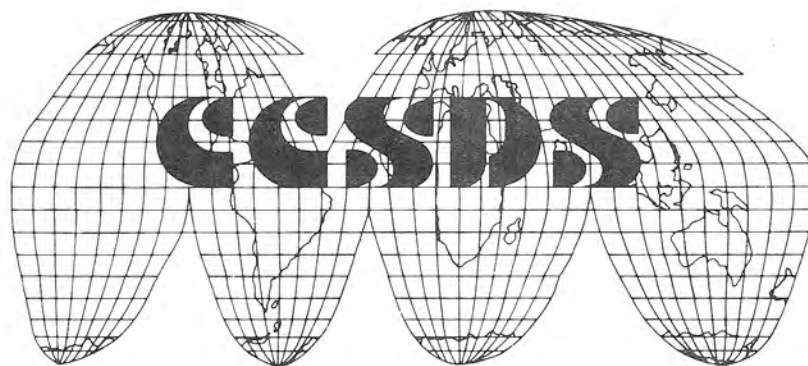
CONSULTATIVE COMMITTEE FOR SPACE DATA SYSTEMS

**RECOMMENDATION FOR SPACE
DATA SYSTEM STANDARDS:**

TELEMETRY CHANNEL CODING

"BLUE BOOK"

MAY 1984



DEDICATION

This document is dedicated to the memory of Mr. Michel F. Pellet of the European Space Agency. His vision and leadership were an inspiration to many technical personnel who contributed to this Recommendation. He will be deeply missed by many members of the CCSDS, and by all people who value the increased international understanding which is achieved through cooperation in the peaceful scientific exploration of space.

CCSDS RECOMMENDATION FOR TELEMETRY CHANNEL CODING

AUTHORITY

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This Recommendation reflects the consensus technical agreement of the following member Agencies of the Consultative Committee for Space Data Systems (CCSDS):

- o Centre National D'Etudes Spatiales (CNES)/France
- o Deutsche Forschungs-u. Versuchsanstalt fuer
Luft und Raumfahrt e.V (DFVLR)/West Germany
- o European Space Agency (ESA)/Europe.
- o Indian Space Research Organization (ISRO)/India
- o Instituto de Pesquisas Espaciais (INPE)/Brazil
- o National Aeronautics and Space Administration (NASA)/USA
- o National Space Development Agency of Japan (NASDA)/Japan.

The following observer Agencies also concur with this Recommendation:

- o Department of Communications, Communications Research
Centre (DOC-CRC)/Canada
- o Institute of Space and Astronautical Science (ISAS)/Japan
- o Radio Research Laboratory (RRL)/Japan

This Recommendation is published and maintained by:

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STATEMENT OF INTENT

The Consultative Committee for Space Data Systems (CCSDS) is an organization officially established by the management of seven member space Agencies. The Committee meets periodically to address data systems problems that are common to all participants, and to formulate sound technical solutions to these problems. Inasmuch as participation in the CCSDS is completely voluntary, the results of Committee actions are termed **RECOMMENDATIONS** and are not considered binding on any Agency.

This RECOMMENDATION is issued by, and represents the consensus of, the CCSDS Plenary body. Agency endorsement of this RECOMMENDATION is entirely voluntary. Endorsement, however, indicates the following understandings:

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 - The STANDARD itself.
 - The anticipated date of initial operational capability.
 - The anticipated duration of operational service.
- o Specific service arrangements shall be made via memoranda of agreement. Neither this RECOMMENDATION nor any ensuing STANDARD is a substitute for a memorandum of agreement.

No later than five years from its date of issuance, this Recommendation will be reviewed by the CCSDS to determine whether it should: (1) remain in effect without change; (2) be changed to reflect the impact of new technologies, new requirements, or new directions; or, (3) be retired or cancelled.

FOREWORD

This document is a technical Recommendation for use in developing telemetry channel coding systems and has been prepared by the Consultative Committee for Space Data Systems (CCSDS). The telemetry channel coding concept described herein is the baseline concept for spacecraft-to-ground data communication within missions that are cross-supported between Agencies of the CCSDS.

This Recommendation establishes a common framework and provides a common basis for the coding schemes used on spacecraft telemetry streams. It allows implementing organizations within each Agency to proceed coherently with the development of compatible derived Standards for the flight and ground systems that are within their cognizance. Derived Agency Standards may implement only a subset of the optional features allowed by the Recommendation and may incorporate features not addressed by the Recommendation.

Through the process of normal evolution, it is expected that expansion, deletion or modification to this document may occur. This Recommendation is therefore subject to CCSDS document management and change control procedures which are defined in Reference [1].

CCSDS RECOMMENDATION FOR TELEMETRY CHANNEL CODING

DOCUMENT CONTROL

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- [1] **"Procedures Manual for the Consultative Committee for Space Data Systems,"** CCSDS, May 1984 or later issue.
- [2] **"Telemetry Coding Standard,"** European Space Agency ESA:TTC-A-03, Issue No. 1, December 1979.
- [3] **"Tracking and Data Relay Satellite System (TDRSS) Users' Guide,"** NASA-Goddard Space Flight Center STDN 101.2, Rev. 4, January, 1980.
- [4] **"Deep Space Network/Flight Project Interface Design Book, Volume II: Proposed DSN Capabilities,"** NASA-Jet Propulsion Laboratory, Document 810-5, Rev. D.
- [5] **"Recommendation for Space Data System Standards: Packet Telemetry,"** CCSDS, Issue-1, May 1984.
- [6] Miller, R. L., et al, **"On the Error Statistics of Viterbi Decoding and the Performance of Concatenated Codes,"** NASA-Jet Propulsion Laboratory Publication 81-9, September 1, 1981.
- [7] Perlman, M., and J. Lee, **"Reed-Solomon Encoders - Conventional vs. Berlekamp's Architecture,"** NASA-Jet Propulsion Laboratory Publication 82-71, December 1, 1982.
- [8] Odenwalder, J. P., et al, **"Hybrid Coding Systems Study,"** Final Report, Contract NAS2-6722, Linkabit Corp., San Diego, Sept. 1972.

1 INTRODUCTION

1.1 PURPOSE

The purpose of this document is to establish a common Recommendation for space telemetry channel coding systems to provide cross-support among missions and facilities of member Agencies of the Consultative Committee for Space Data Systems (CCSDS.) In addition, it provides focussing for the development of multi-mission support capabilities within the respective agencies to eliminate the need for arbitrary, unique capabilities for each mission.

Telemetry channel coding is a method by which data can be sent from a source to a destination by processing data so that distinct messages are created which are easily distinguishable from one another. This allows reconstruction of the data with low error probability, thus improving the performance of the channel.

This document was prepared by the CCSDS primarily for the purpose of facilitating the cross-support concept through standardizing key items of data systems compatibility. While the CCSDS has no power of enforcement, it is expected that this recommendation will be incorporated into each respective Agency's data systems standards, and through them, will apply to all missions that wish to utilize telemetry channel coding for cross-support.

1.2 SCOPE

Several space telemetry channel coding schemes are described in this document. The characteristics of the codes are specified only to the extent necessary to ensure interoperability and cross-support. The specification does not attempt to quantify the relative coding gain or the merits of each approach discussed, nor the design requirements for encoders or decoders. Some performance information is included in Annex A, Rationale.

This recommendation does not require that coding be used on all cross supported missions. However, for those planning to use coding, the recommended codes to be used are those described in this document.

The rate 1/2 convolutional code recommended for cross-support is described in Section 2, "Convolutional Coding." Depending on performance requirements, this code alone may be satisfactory.

Users of the NASA Tracking and Data Relay Satellite System (TDRSS) may be required to use periodic convolutional interleaving in addition to the convolutional code above. This approach is

In accordance with modern data communications practice, spacecraft data fields are often grouped into 8-bit "words" which conform to the above convention. Throughout this Recommendation, the following nomenclature is used to describe this grouping:

"8-bit word" = "Octet"

2 CONVOLUTIONAL CODING

The basic code selected for cross-support is a rate $1/2$, constraint-length 7 convolutional code. It may be used alone, as described in this section, or in conjunction with enhancements described in the following sections. While slightly different conventions of this code, currently in use by some member Agencies, may continue to be supported for an interim period, it is the recommendation of the CCSDS to universally adopt the single convention described herein.

This recommendation is a non-systematic code and a specific decoding procedure, with the following characteristics:^{1,2}

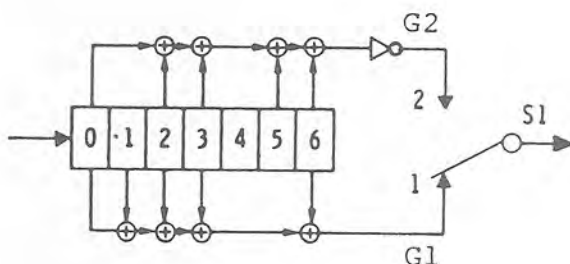
- a. Nomenclature: Convolutional code with maximum-likelihood (Viterbi) decoding.
- b. Code rate: $1/2$ bit per symbol
- c. Constraint length: 7 bits
- d. Connection vectors: $G1 = 1111001$; $G2 = 1011011$
- e. Phase relationship: $G1$ is associated with first Symbol
- f. Symbol inversion: On output path of $G2$

An encoder block diagram with the recommended convention is shown in Fig. 2-1.

It is recommended that soft bit decisions with at least 3-bit quantization be used whenever constraints (such as location of decoder) permit.

Footnote 1: The following upper bounds to the data rates for spacecraft telemetry reception may exist because of symbol synchronizer or decoder limitations:

- (a) ESA: Maximum symbol rate is obtained at a data rate of 1 Mb/s NRZ-L, or 500 kb/s split-phase. (Ref. [2])
- (b) NASA-GSFC: Maximum symbol rate is obtained at a data rate of 3 Mb/s NRZ. (Ref. [3])
- (c) NASA-JPL: Current maximum symbol rate is 250 ksymbols/s. A planned upgrade will be decoder-limited at 250 kb/s. (Ref. [4])



1.  = Single bit delay
2. For every input bit, two symbols are generated by completion of a cycle for S1: Pos'n 1, Pos'n 2.
3. S1 is in the position shown (1) for the first symbol associated with an incoming bit.
4. $\oplus$ = modulo-2 adder.
5.  = inverter.

Figure 2-1. Convolutional Encoder Block Diagram.

Footnote 2: When suppressed-carrier modulation systems are used, NRZ-M or NRZ-L may be used as a modulating waveform. If the user contemplates conversion of his modulating waveform from NRZ-L to NRZ-M, such conversion should be performed on-board at the input to the convolutional encoder. Correspondingly, the conversion on the ground from NRZ-M to NRZ-L should be performed at the output of the convolutional decoder. This avoids unnecessary link performance loss and is shown in Fig. A-1.

CAUTION: When a fixed pattern in the symbol stream is used to provide node synchronization for the Viterbi decoder, care must be taken to account for any translation of the pattern due to the modulating waveform conversion.

3 CONVOLUTIONAL CODING WITH INTERLEAVING FOR TRACKING AND DATA RELAY SATELLITE OPERATIONS

3.1 INTRODUCTION

Users of the TDRSS S-band Single Access (SSA) Channel, where the channel symbol rate exceeds 300 ks/s, will be required to employ interleaving in conjunction with the convolutional code which has been described in Section 2. Users are cautioned that if such interleaving is not used under these conditions, the Goddard Space Flight Center Networks Directorate does not guarantee the specified performance and will not be obligated to troubleshoot the system in case of problems. (Ref. [3])

It should be noted that this interleaving is totally separate and distinct from the interleaving used in conjunction with the Reed-Solomon code described in Section 4.

3.2 DESCRIPTION

The type of interleaving required is called "Periodic Convolutional Interleaving" and is specified in Appendix J of the TDRSS Users's Guide. (Ref. [3])

3.3 BYPASS CAPABILITY

A TDRSS-compatible spacecraft using the Periodic Convolutional Interleaving specified in this section must be capable of bypassing its Periodic Convolutional Interleaver in the event direct support from a non-TDRSS ground tracking station is desired. This is because the interference that this interleaving is designed to protect against is not harmful in this configuration, and moreover, the necessary de-interleavers do not exist at these ground stations.

4 REED-SOLOMON CODING

4.1 INTRODUCTION

While a convolutional code provides good forward error correction capability in a gaussian noise channel, significant additional improvement (particularly to correct bursts of errors from the Viterbi decoder) can be obtained by concatenating a Reed-Solomon (R-S) code with the convolutional code. The Reed-Solomon code forms the outer code, while the convolutional code is the inner code. The overhead associated with the Reed-Solomon code is comparatively low, and the improvement in the error performance can often provide the nearly-error-free channel required to support efficient automated ground handling of space mission telemetry.

The user is cautioned that the R-S outer code described in this section is not intended for use except when concatenated with the inner convolutional code described in Section 2.

The TDRS System does not furnish any Reed-Solomon decoding services.

4.2 SPECIFICATION

The parameters of the selected Reed-Solomon code are as follows:

- (a) $J = 8$ bits per R-S symbol
- (b) $E = 16$ R-S symbol error correction capability within a Reed-Solomon codeword.
- (c) General characteristics of Reed-Solomon codes
 - 1. J , E , and I , the depth of interleaving, are independent parameters.
 - 2. $n = 2^J - 1 = 255$ symbols per R-S codeword.
 - 3. $2E$ is the number of R-S symbols among n symbols of an R-S codeword representing checks.
 - 4. $k = n - 2E$ is the number of R-S symbols among n R-S symbols of an R-S codeword representing information.
- (d) Field generator polynomial:

$$F(x) = x^8 + x^7 + x^2 + x + 1$$

over $GF(2)$

(e) Code generator polynomial:

$$g(x) = \prod_{j=1}^{143} (x - a^{11j}) = \sum_{i=0}^{32} G_i x^i$$

over $GF(2^8)$,
where $F(a) = 0$.

It should be recognized that a^{11} is a primitive element in $GF(2^8)$ and that $F(x)$ and $g(x)$ characterize a (255,223) Reed-Solomon code.

(f) The selected code is a systematic code. This results in a systematic codeblock.

(g) Symbol Interleaving

Symbol interleaving is accomplished in a manner functionally described with the aid of Fig. 4-1. (It should be noted that this functional description does not necessarily correspond to the physical implementation of an encoder.)

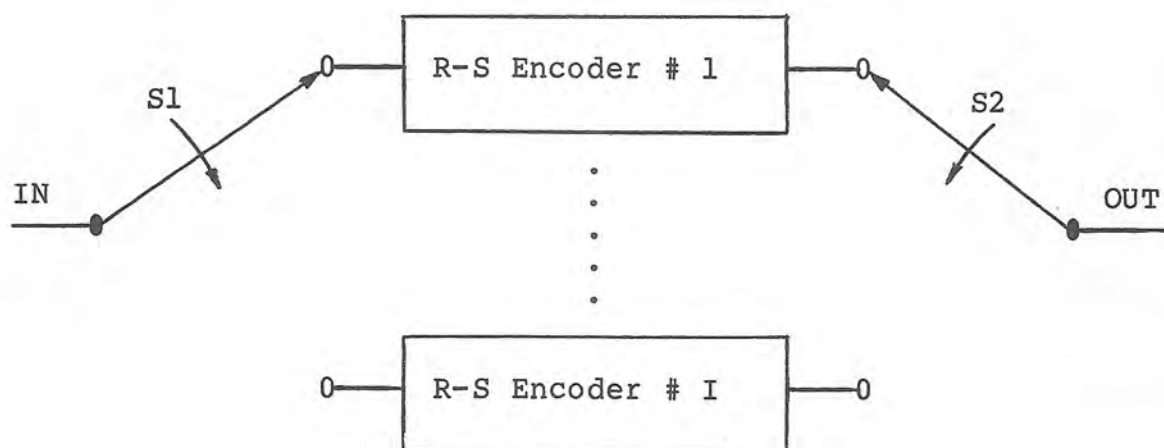


Fig. 4-1. Functional Representation of R-S Interleaving

Data bits to be encoded into a single Reed-Solomon Codeblock enter at the port labeled "IN." Switches S1 and S2 are synchronized together and advance from encoder to encoder in the sequence 1,2, ... I, 1 ... , spending one R-S symbol time (8 bits) in each position.

One codeblock will be formed from 223I R-S symbols entering "IN." In this functional representation, a space of 32I R-S symbols in duration is required between each entering set of 223I R-S information symbols.

Due to the action of S1, each encoder accepts 223 of these symbols, each symbol spaced I symbols apart (in the original stream.) These 223 symbols are passed directly to the output of each encoder. The synchronized action of S2 reassembles the symbols at the port labeled "OUT" in the same way as they entered at "IN."

Following this, each encoder outputs its 32 check symbols, one symbol at a time, as it is sampled in sequence by S2.

If, for $I=5$, the original symbol stream is

1 5 1 5 1 5
d ... d d ... d d ... d [32 x 5 spaces]
1 1 2 2 223 223

then the output is the same sequence followed by the [32x5] parity symbols as shown below:

$$p_1^1 \dots p_1^5 \dots p_{32}^1 \dots p_{32}^5$$

where

$$d_{11}^{(i)} d_{12}^{(i)} \dots d_{223}^{(i)} p_1^{(i)} \dots p_{32}^{(i)}$$

is the R-S codeword produced by the i^{th} encoder. If q virtual fill symbols are used in each codeword, then replace 223 by $(223 - q)$ in the above discussion.

With this method of interleaving, the original kI consecutive information symbols that entered the encoder appear unchanged at the output of the encoder with 2EI R-S check symbols appended.

The recommended value of interleaving depth is $I=5$, but $I=1$ is permitted. (See Footnote 3.)

(h) Maximum Code Block Length

The maximum code block length, in R-S symbols, is given by:

$$L_{\max} = nI = (2^J - 1)I = 255I$$

Footnote 3: Users of TDRSS are cautioned that, under some special RFI circumstances, additional measures may have to be employed to obtain a required performance. One such approach may be to utilize a Reed-Solomon code with a different depth of interleaving. In addition to I=1 and I=5, ESA will support I=8.

(i) Shortened Code Block Length⁴

A shortened code block length may be used to accommodate frame lengths smaller than the maximum. However, since the Reed-Solomon code is a block code, the decoder must always operate on a full block basis. To achieve a full code block, "virtual fill" must be added to make up the difference between the shortened block and the maximum code block length. The characteristics and limitations on virtual fill are covered in paragraph (j.) Since the virtual fill is not transmitted, both encoder and decoder must be set to insert it with the proper length for the encoding and decoding processes to be carried out properly.

When an encoder (initially cleared at the start of a block) receives $kI-Q$ symbols representing information, (where Q , representing fill, is a multiple of I , and is less than kI), $2EI$ check symbols are computed over kI symbols, of which the leading Q symbols are treated as all-zero symbols. A $(nI-Q, kI-Q)$ shortened codeblock results where the leading Q symbols (all zeros) are neither entered into the encoder nor transmitted.

(j) Partitioning and Virtual Fill

The codeblock is partitioned as shown in Fig. 4-2.

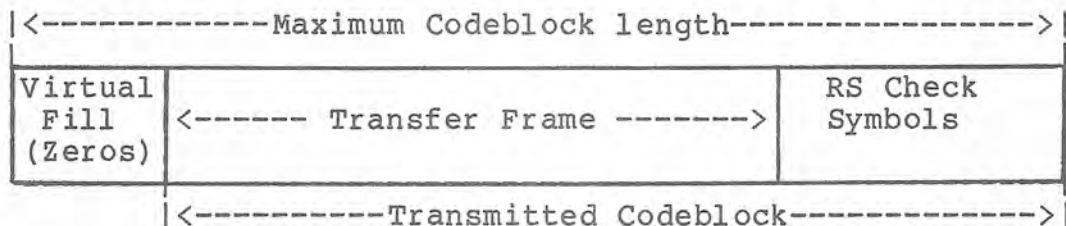


Fig. 4-2 Codeblock Partitioning

The Reed-Solomon Check Symbols consist of the trailing $2EI$ symbols ($2EIJ$ bits) of the codeblock.

The Transfer Frame is defined by the CCSDS Recommendation for Packet Telemetry (Reference [5]). For $I=5$, it has a length of 8920 bits, which includes the 32-bit R-S codeblock marker used to synchronize the Reed-Solomon codeblock.

Footnote 4: It should be noted that shortening the block length in this way changes the overall performance to a degree dependent on the amount of virtual fill used. Since it incorporates no virtual fill, the Packet Telemetry transfer frame length recommended in Reference [5] allows full performance.

The Transmitted Codeblock consists of what is actually transmitted on the space telemetry channel (i.e., it consists of all bits transmitted from the beginning of one R-S codeblock marker to the beginning of the next R-S codeblock marker.) For $I=5$ and with no virtual fill, this is 10,200 bits.

If used, virtual fill shall:

- a) consist of all zeros,
- b) not be transmitted.
- c) not change in length during a tracking pass.
- d) be inserted, prior to decoding, only at the beginning of the codeblock (i.e., before the transmitted frame sync word.)
- e) be inserted only in integer multiples of $8I$ bits.

(k) Symbol representation and ordering for transmission

Each 8-bit Reed-Solomon symbol is an element of the finite field $GF(256)$. Since $GF(256)$ is a vector space of dimension 8 over the binary field $GF(2)$, the actual 8-bit representation of a symbol is a function of the particular basis that is chosen. One basis for $GF(256)$ over $GF(2)$ is the set $\{1, a^1, a^2, \dots, a^7\}$. This means that any element b of $GF(256)$ has a representation of the form

$$b = u_7 a^7 + u_6 a^6 + \dots + u_1 a^1 + u_0 1$$

where each u_i is either a zero or a one.

There is another basis $\{l_0, l_1, \dots, l_7\}$ called the "dual basis" to the above basis. It has the property that

$$\text{Tr}(l_i a^j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

for each $j = 0, 1, \dots, 7$. The function $\text{Tr}(z)$, called the "trace," is defined by

$$\text{Tr}(z) = \sum_{k=0}^7 z^{2^k}$$

for each element z of $GF(256)$. Each Reed-Solomon symbol can also be represented as

$$b = z_0 l_0 + z_1 l_1 + \dots + z_7 l_7$$

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where each z_i is either a zero or a one. The representation recommended in this document is the dual basis eight bit string $z_0, z_1, \dots, z_7$, transmitted in that order (with z_0 first.) The relationship between the two representations is given by the two equations

$$[z_0, \dots, z_7] = [u_7, \dots, u_0]$$

and

$$[u_7, \dots, u_0] = [z_0, \dots, z_7]$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Further information relating the dual basis (Berlekamp) and conventional representations is given in Annex B. Also included is a recommended scheme for permitting the symbols generated in a conventional encoder to be transformed to meet the symbol representation required by this document.

(1) Synchronization⁵

Codeblock synchronization of the Reed-Solomon decoder is achieved by synchronization of the 32-bit R-S codeblock marker. This marker may be supplied as part of the Packet Telemetry Transfer Frame (Ref. [5]) or it may be supplied as part of the R-S encoding layer. The bit pattern for the marker is:
(Currently under study.)

(m) Ambiguity Resolution

The ambiguity between true and complemented data must be resolved so that only true data is provided to the Reed-Solomon decoder. Data in NRZ-L form is normally resolved using the 32-bit R-S codeblock marker, while NRZ-M data is self-resolving.

Footnote 5: CAUTION! If the Packet Telemetry Transfer Frame Format defined in Reference [5] is NOT applied to the data space of the R-S codeblock, then the user is cautioned that under certain repeating data conditions the sync marker pattern may be regenerated in the check symbol field of the codeblock and this may lead to synchronization difficulties.

ANNEX A:
RATIONALE

Purpose:

While the body of this document treats the technical specifications for the recommended coding systems, this annex provides the rationale for the choices of the coding types and their specific parameters.

Status:

(THIS ANNEX IS NOT PART OF THE RECOMMENDATION)

1 INTRODUCTION

Channel coding is a method by which data can be sent from a source to a destination by processing data so that distinct messages are easily distinguishable from one another. This allows reconstruction of the data with low error probability.

In spacecraft, the data source is usually digital, with the data represented as a string of zeroes and ones. A channel encoder (or simply "encoder") is then a device that takes this string of binary data and produces a modulating waveform as output. If the channel code is chosen correctly for the particular channel in question, then a properly designed decoder will be able to reconstruct the original binary data even if the waveforms have been corrupted by channel noise. If the characteristics of the channel are well understood, and an appropriate coding scheme is chosen, then channel coding provides a higher overall data throughput at the same overall quality (bit error rate) as uncoded transmission - but with less energy expended per information bit. Equivalently, channel coding allows a lower overall bit error rate than the uncoded system using the same energy per information bit.

There are three other benefits that may be expected from coding. First, the resulting "clean" channel can benefit the transmission of compressed data. The purpose of data compression schemes is to map a large amount of input data into a smaller number of bits. This is done by removing redundant data. Since the number of bits that are actually transmitted is often much lower than in an uncompressed system and each bit carries more information, compressed data are more sensitive to errors.

Second, a low bit error rate is also required when adaptive telemetry is used. In adaptive telemetry, information on how various ground processors should treat the encoded data is included as part of the data itself. An error in these bits could result in improper handling of subsequent data and the possible loss of much information.

Third, low error probability telemetry may allow a certain amount of unattended mission operations. This is principally because the operations systems will know that any anomalies detected in the downlink data are extremely likely to be real and not caused by channel errors. Thus, operators may not be required to try to distinguish erroneous data from genuine spacecraft anomalies.

In a typical space channel, the principal signal degradations are due to the loss of signal energy with distance, and to the thermal noise in the receiving system. The codes described in this document can usually provide good communication over this channel. An additional degradation, caused by interference from earth-based pulse radars, may occur for users of TDRSS. Such users may consider adding periodic convolutional interleaving (PCI) to their coding system; in this case, they should carefully analyze the effects of the PCI on their systems.

If interagency cross support requires one agency to decode the telemetry of another, then the codes recommended in this document should be used. A block diagram of the recommended coding system appears in Figure A-1.

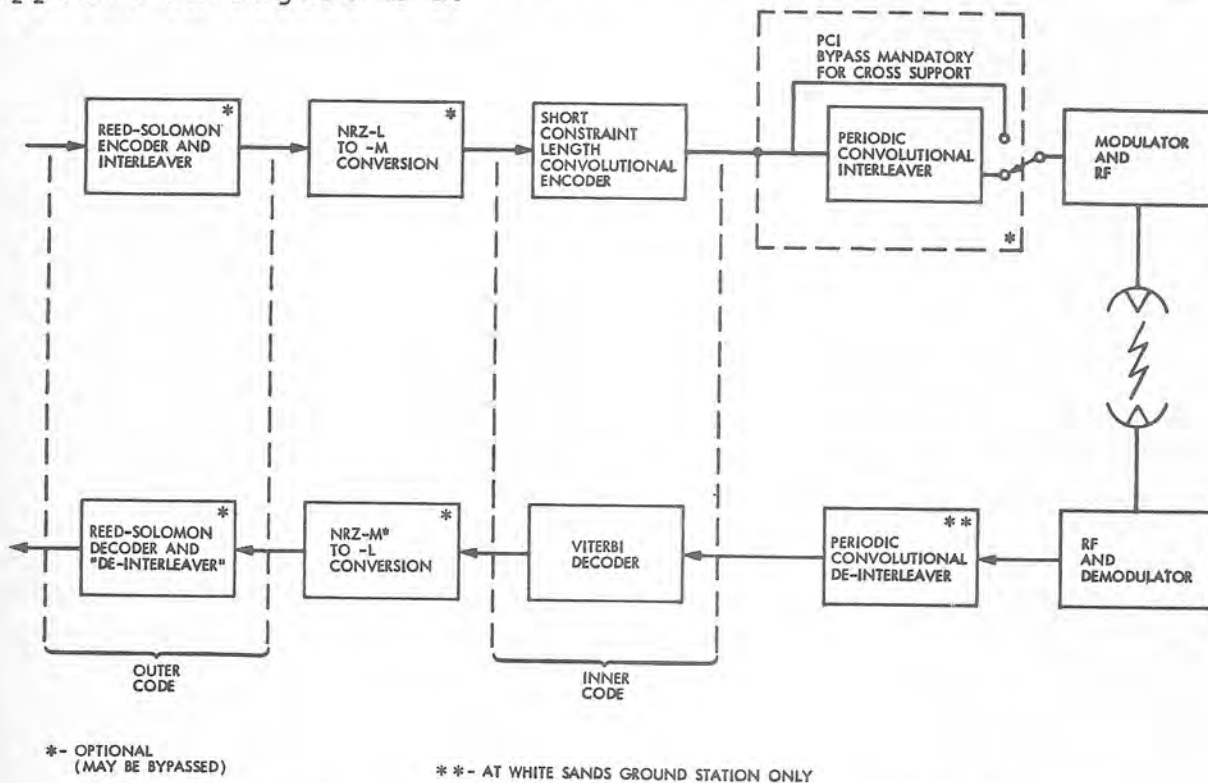


Figure A-1. Coding System Block Diagram.

The relative performance of the various codes in a Gaussian channel is shown in Figure A-2 (from Ref. [6].) Here, the input is constrained to be chosen from between two levels, because biphase modulation is assumed throughout this recommendation. These performance data were obtained by software simulation and assume that there are no synchronization losses. The channel symbol errors were assumed to be independent. This is a good assumption for the deep space channel. Also, infinite interleaving was assumed in the Reed-Solomon code. It is clear from the figure that the convolutional code offers a coding gain of about 5.5 dB over an uncoded system at a decoded bit error rate of 10^{-5} . The use of the outer Reed-Solomon code results in an additional 2.0 dB of coding gain. Note that figure A-2 does not necessarily represent the performance of the TDRSS channel.

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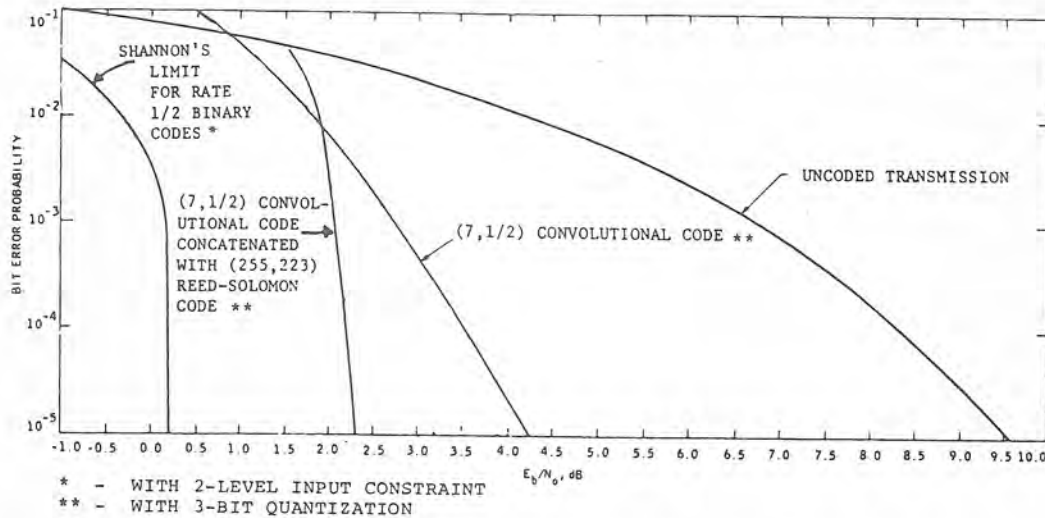


Figure A-2. Performance of various codes in a Gaussian channel.

These codes are included in the recommendation because they represent state-of-the-art coding technology and provide substantial coding gain over an uncoded system. They have already been incorporated, or are planned to be incorporated, into missions of member Agencies of the CCSDS.

The next three sections explain the choice of the codes and the parameters of each code in more detail.

2 THE CONVOLUTIONAL CODE

A rate 1/2, constraint length 7 convolutional code with Viterbi (maximum likelihood) decoding is already a standard for both NASA and ESA. It has been used in several missions and has demonstrated the expected coding gain.

The encoder for this code is extremely simple. It consists of a shift register of length six and some exclusive OR gates that implement the two parity checks. The two checks are then multiplexed into one line. This means that the encoder can be made small and that it dissipates very little power. These are good attributes for spacecraft hardware.

It has been customary to invert one or the other parity check in the encoder. This is to ensure that there are sufficient transitions in the channel stream for the symbol synchronizer to work in the case of a steady state (all zeroes or all ones) input to the encoder.

Historically, ESA, NASA-GSFC and NASA-JPL each have used a different ordering of the two parity checks or has used a different check inverted. Performance is not affected by these minor differences. While interim cross support of these different conventions may require minor differences in ground station equipment, all Agencies are encouraged to adopt for all facilities the single convention described in this document, which is the NASA-GSFC convention.

3 PERIODIC CONVOLUTIONAL INTERLEAVING

Low earth-orbiting spacecraft sending telemetry to the ground using the services of the TDRSS S-band Single Access (SSA) channel when symbol rates exceed 300 ks/s may experience pulsed radio interference which is expected to severely degrade the link performance during certain portions of the user orbit. In order to be able to maintain specified performance on this link at all times, the user satellite must employ an interleaving technique in conjunction with the convolutional coding and must increase the effective isotropic radiated power (EIRP). These techniques will insure that no more than one of the dependent symbol errors due to a single RFI pulse is within the path memory length of the decoder at any given time, and that the signal energy has been increased sufficiently to offset the increased symbol error probability.

The interleaving parameters have been selected to achieve this goal for a particular worst-case pulse interference signature and the maximum symbol rate (6 Ms/s) of the SSA channel. De-interleaving must take place before convolutional decoding, and therefore is accomplished at the White Sands Ground Terminal.

4 THE REED-SOLOMON CODE.

Due to the nature of Viterbi decoding, the decoded bit errors of the (7, 1/2) convolutional code tend to clump together in bursts. For this reason, in a concatenated coding system that uses a convolutional inner code, the outer code should be tailored to a burst error environment.

The code that is recommended as the outer code is a (255, 223) Reed-Solomon code. This code is a non-binary code. Each member of its coding alphabet is one of 256 elements of a finite field rather than a zero or a one. A string of eight bits is used to represent elements in the field so that the output of the encoder still looks like binary data.

Reed-Solomon codes are block codes. This means that a fixed block of input data is processed into a fixed block of output data. In the case of the (255, 223) code, 223 Reed-Solomon input symbols (each eight bits long) are encoded into 255 output symbols. The Reed-Solomon code in the recommendation is systematic. This means that some portion of the codeword contains the input data in unaltered form. In this case, the first 223 symbols are the input data. The Reed-Solomon decoder almost always knows when there are too many errors to correct a word. In the event this happens, the decoder can inform the user of this fact.

A Reed-Solomon symbol size of eight bits was chosen because the decoders for larger symbol sizes would be difficult to implement with current technology. This choice forces the longest codeword length to be 255 symbols. A 16 R-S symbol error correction capability was chosen as this was shown to have the best performance

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when concatenated with the $(7, 1/2)$ convolutional inner code (Ref. [8].) Since two check symbols are required for each error to be corrected, this results in a total of 32 check symbols and 223 information symbols per codeword.

The $(255, 223)$ Reed-Solomon code is capable of correcting up to 16 Reed-Solomon symbol errors in each codeword. Since each symbol is actually eight bits, this means that the code can correct up to 16 short bursts of error due to the inner convolutional decoder.

In addition, the Reed-Solomon codewords can be interleaved on a symbol basis before being convolutionally encoded. Since this separates the symbols in a codeword, it becomes less likely that a burst from the Viterbi decoder disturbs more than one Reed-Solomon symbol in any one codeword. This improves the performance of the Reed-Solomon code. An interleaving depth of five was chosen for two reasons. A depth of five results in performance that is virtually indistinguishable from a depth of infinity. Also, a depth of five results in a frame length (a set of five codewords) that is a good compromise considering ease of handling, data outages (quality, quantity and continuity,) and frame sync rate.

The same encoding and decoding hardware can implement a shortened $(n, n-32)$ Reed-Solomon code where $n=33, 34, \dots, 254$. This is accomplished by assuming that the remaining symbols are fixed: in the case of the recommendation, they are assumed to be all zero. This virtual zero fill allows the frame length to be tailored, if necessary, to suit a particular mission or situation.

The method currently recommended for synchronizing the codeblock is by synchronization of the transfer frame which contains a frame synchronization marker of 32 bits. However, advanced approaches being studied (e.g., self-synchronizing Reed-Solomon codes) may enable these two functions to be separately synchronized in the future.

The Reed-Solomon code, like the convolutional code of Section 2, is transparent. This means that if the channel symbols have been inverted somewhere along the line, the decoders will still operate. The result will be the complement of the original data. The Reed-Solomon code loses its transparency if virtual zero fill is used. For this reason it is mandatory that the sense of the data (i.e., true or complemented) be resolved before Reed-Solomon decoding.

The two polynomials that define the Reed-Solomon code (Sec. 4.2 (d) and (e)) were chosen to minimize the encoder hardware. The code generator polynomial is a palindrome (self-reciprocal polynomial) so that only half as many multipliers are required in the encoder circuit. The particular primitive element, "a", (and hence the field generator polynomial,) was chosen to make these multipliers as simple as possible. An encoder using the "dual basis" representation requires for implementation only a small number of integrated circuits or a single VLSI chip.

ANNEX B:

**TRANSFORMATION BETWEEN BERLEKAMP
AND CONVENTIONAL REPRESENTATIONS**

Purpose:

This annex provides information to assist users of the Reed-Solomon code in this Recommendation to transform between the Berlekamp (dual basis) and Conventional representations. In addition, it shows where transformations are made to allow a conventional encoder to produce the dual basis representation on which the Recommendation is based.

Status:

(THIS ANNEX IS NOT PART OF THE RECOMMENDATION)

Referring to Figure B-1, it can be seen that information symbols I entering, and check symbols C emanating from the Berlekamp RS encoder are interpreted as

$$[z_0, z_1, \dots, z_7]$$

where the components z_i are coefficients of l_i , respectively:

$$z_0 l_0 + z_1 l_1 + \dots + z_7 l_7$$

Information symbols I' entering and check symbols C' emanating from the conventional RS encoder are interpreted as

$$[u_7, u_6, \dots, u_0]$$

where the components u_j are coefficients of a^j , respectively.

$$u_7 a^7 + u_6 a^6 + \dots + u_0$$

A pre- and post-transformation is required when employing a conventional RS encoder.

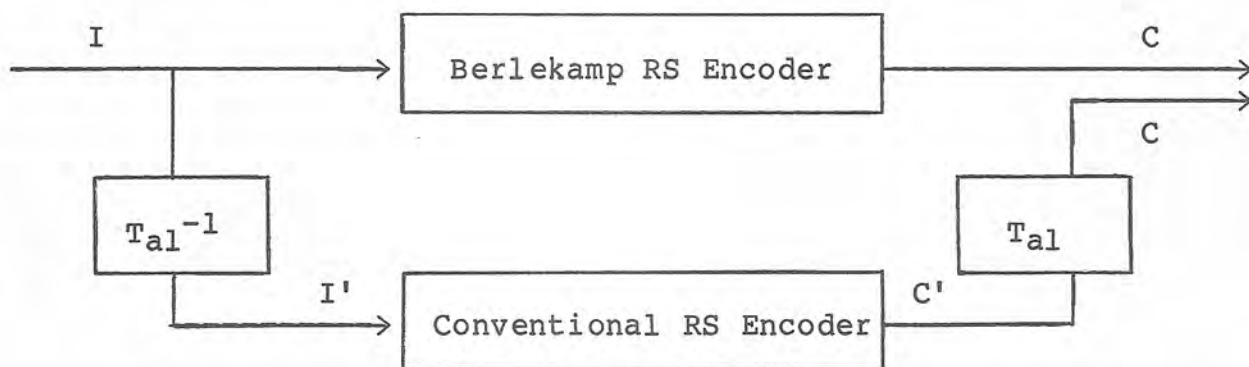


Fig. B-1. Transformational Equivalence

Conventional and Berlekamp types of (255,223) Reed-Solomon encoders are assumed to have the same self-reciprocal generator polynomial whose coefficients appear in paragraph 4.2 (d) and (e). The representation of symbols associated with the conventional encoder are the polynomials in "a" appearing in Table B-1, below. Corresponding to each polynomial in "a" is the representation in the dual basis of symbols associated with the Berlekamp type encoder.

Given

$$a^i = u_7 a^7 + u_6 a^6 + \dots + u_0$$

where $0 \leq i < 255$

and a^* denotes the zero polynomial, $u_7, u_6, \dots = 0$,
the corresponding element is

$$z = z_0 1_0 + z_1 1_1 + \dots + z_7 1_7$$

where $[z_0, z_1, \dots, z_7] = [u_7, u_6, \dots, u_0] T_{a1}$

and

$$T_{a1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

Row 1, row 2, ..., and row 8 in T_{a1} are representations in the dual basis of a^7 ($1_0 \dots 0$), a^6 ($01_0 \dots 0$), ..., and a^0 ($00 \dots 01$), respectively.

The inverse of T_{a1} is

$$T_{a1}^{-1} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Row 1, row 2, ..., and row 8 in T_{a1}^{-1} are polynomials in "a" corresponding to 1_0 ($1_0 \dots 0$), 1_1 ($01_0 \dots 0$), ..., and 1_7 ($00, \dots 01$), respectively. Thus,

$$[z_0, z_1, \dots, z_7] T_{a1}^{-1} = [u_7, u_6, \dots, u_0]$$

Example 1:

Given information symbol I, $[z_0, z_1, \dots, z_7] = 10111001$

then

$$[10111001] T_{a1}^{-1} = [u_7, u_6, \dots, u_0] = 00101010 = I'$$

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Note that the arithmetic operations are reduced modulo 2. Also,

$$[z_0, z_1, \dots, z_7] = 10111001$$

and $[u_7, u_6, \dots, u_0] = 00101010 \text{ (a}^{213}\text{)}$

are corresponding entries in Table B-1.

Example 2:

Given check symbol C' ,

$$[a_7, a_6, \dots, a_0] = 01011001 \text{ (a}^{152}\text{)}$$

Then,

$$[01011001] \begin{bmatrix} 10001101 \\ 11101111 \\ 11101100 \\ 10000110 \\ 11111010 \\ 10011001 \\ 10101111 \\ 01111011 \end{bmatrix}$$

$$= [z_0, z_1, \dots, z_7] = 11101000 = C$$

Table B-1. Equivalence of Representations.¹

P O W E R	POLY IN ALPHA	<i>L</i> 01234567	P O W E R	POLY IN ALPHA	<i>L</i> 01234567
=====			=====		
*	00000000	00000000	31	11001101	01111010
0	00000001	01111011	32	00011101	10011110
1	00000010	10101111	33	00111010	00111111
2	00000100	10011001	34	01110100	00011100
3	00001000	11111010	35	11101000	01110100
4	00010000	10000110	36	01010111	00100100
5	00100000	11101100	37	10101110	10101101
6	01000000	11101111	38	11011011	11001010
7	10000000	10001101	39	00110001	00010001
8	10000111	11000000	40	01100010	10101100
9	10001001	00001100	41	11000100	11111011
10	10010101	11101001	42	00001111	10110111
11	10101101	01111001	43	00011110	01001010
12	11011101	11111100	44	00111100	00001001
13	00111101	01110010	45	01111000	01111111
14	01111010	11010000	46	11110000	00001000
15	11110100	10010001	47	01100111	01001110
16	01101111	10110100	48	11001110	10101110
17	11011110	00101000	49	00011011	10101000
18	00111011	01000100	50	00110110	01011100
19	01110110	10110011	51	01101100	01100000
20	11101100	11101101	52	11011000	00011110
21	01011111	11011110	53	00110111	00100111
22	10111110	00101011	54	01101110	11001111
23	11111011	00100110	55	11011100	10000111
24	01110001	11111110	56	00111111	11011101
25	11100010	00100001	57	01111110	01001001
26	01000011	00111011	58	11111100	01101011
27	10000110	10111011	59	01111111	00110010
28	10001011	10100011	60	11111110	11000100
29	10010001	01110000	61	01111011	10101011
30	10100101	10000011	62	11110110	00111110

Footnote 1: From Table 4 of Ref. [7].

Note: Coefficients of the "Polynomial in Alpha" column are the same as the "Polynomial in a" as used in this recommendation, and are listed in descending powers of a, starting with a⁷.

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Table B-1, Cont'd.

P O W E R	POLY IN ALPHA	<i>l</i> 01234567	P O W E R	POLY IN ALPHA	<i>l</i> 01234567
=====			=====		
63	01101011	00101101	95	10111010	10110010
64	11010110	11010010	96	11110011	11011100
65	00101011	11000010	97	01100001	01111000
66	01010110	01011111	98	11000010	11001101
67	10101100	00000010	99	00000011	11010100
68	11011111	01010011	100	00000110	00110110
69	00111001	11101011	101	00001100	01100011
70	01110010	00101010	102	00011000	01111100
71	11100100	00010111	103	00110000	01101010
72	01001111	01011000	104	01100000	00000011
73	10011110	11000111	105	11000000	01100010
74	10111011	11001001	106	00000111	01001101
75	11110001	01110011	107	00001110	11001100
76	01100101	11100001	108	00011100	11100101
77	11001010	00110111	109	00111000	10010000
78	00010011	01010010	110	01110000	10000101
79	00100110	11011010	111	11100000	10001110
80	01001100	10001100	112	01000111	10100010
81	10011000	11110001	113	10001110	01000001
82	10110111	10101010	114	10011011	00100101
83	11101001	00001111	115	10110001	10011100
84	01010101	10001011	116	11100101	01101100
85	10101010	00110100	117	01001101	11110111
86	11010011	00110000	118	10011010	01011110
87	00100001	10010111	119	10110011	00110011
88	01000010	01000000	120	11100001	11110101
89	10000100	00010100	121	01000101	00001101
90	10001111	00111010	122	10001010	11011000
91	10011001	10001010	123	10010011	11011111
92	10110101	00000101	124	10100001	00011010
93	11101101	10010110	125	11000101	10000000
94	01011101	01110001	126	00001101	00011000

Table B-1, Cont'd.

P O W E R	POLY IN ALPHA	<i>l</i> 01234567	P O W E R	POLY IN ALPHA	<i>l</i> 01234567
=====			=====		
127	00011010	11010011	159	10000101	01101111
128	00110100	11110011	160	10001101	10010101
129	01101000	11111001	161	10011101	00010011
130	11010000	11100100	162	10111101	11111111
131	00100111	10100001	163	11111101	<u>00010000</u>
132	01001110	00100011	164	01111101	10011101
133	10011100	01101000	165	11111010	01011101
134	10111111	01010000	166	01110011	01010001
135	11111001	10001001	167	11100110	10111000
136	01110101	01100111	168	01001011	11000001
137	11101010	11011011	169	10010110	00111101
138	01010011	10111101	170	10101011	01001111
139	10100110	01010111	171	11010001	10011111
140	11001011	01001100	172	00100101	00001110
141	00010001	11111101	173	01001010	10111010
142	00100010	01000011	174	10010100	10010010
143	01000100	01110110	175	10101111	11010110
144	10001000	01110111	176	11011001	01100101
145	10010111	01000110	177	00110101	10001000
146	10101001	11100000	178	01101010	01010110
147	11010101	00000110	179	11010100	01111101
148	00101101	11110100	180	00101111	01011011
149	01011010	00111100	181	01011110	10100101
150	10110100	01111110	182	10111100	10000100
151	11101111	00111001	183	11111111	10111111
152	01011001	11101000	184	01111001	<u>00000100</u>
153	10110010	01001000	185	11110010	10100111
154	11100011	01011010	186	01100011	11010111
155	01000001	10010100	187	11000110	01010100
156	10000010	00100010	188	00001011	00101110
157	10000011	01011001	189	00010110	10110000
158	10000001	11110110	190	00101100	10001111

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Table B-1, Concluded

P O W E R	POLY IN ALPHA	<i>L</i> 01234567	P O W E R	POLY IN ALPHA	<i>L</i> 01234567
=====			=====		
191	01011000	10010011	223	01100100	10011010
192	10110000	11100111	224	11001000	10011000
193	11100111	11000011	225	00010111	11001011
194	01001001	01101110	226	00101110	<u>00100000</u>
195	10010010	10100100	227	01011100	00001010
196	10100011	10110101	228	10111000	00011101
197	11000001	00011001	229	11110111	01000101
198	00000101	11100010	230	01101001	10000010
199	00001010	01010101	231	11010010	01001011
200	00010100	00011111	232	00100011	00111000
201	00101000	00010110	233	01000110	11011001
202	01010000	01101001	234	10001100	11101110
203	10100000	01100001	235	10011111	10111100
204	11000111	00101111	236	10111001	01100110
205	00001001	10000001	237	11110101	11101010
206	00010010	00101001	238	01101101	00011011
207	00100100	01101010	239	11011010	10110001
208	01001000	00010101	240	00110011	10111110
209	10010000	00001011	241	01100110	00110101
210	10100111	00101100	242	11001100	<u>00000001</u>
211	11001001	11100011	243	00011111	00110001
212	00010101	01100100	244	00111110	10100110
213	00101010	10111001	245	01111100	11100110
214	01010100	11110000	246	11111000	11110010
215	10101000	10011011	247	01110111	11001000
216	11010111	10101001	248	11101110	01000010
217	00101001	01101101	249	01011011	01000111
218	01010010	11000110	250	10110110	11010001
219	10100100	11111000	251	11101011	10100000
220	11001111	11010101	252	01010001	00010010
221	00011001	00000111	253	10100010	11001110
222	00110010	11000101	254	11000011	10110110

ANNEX C:

EXPANSION OF REED-SOLOMON COEFFICIENTS

Purpose:

While the equations given in the Reed-Solomon Coding Section of this Recommendation are fully specifying, this annex provides additional assistance for those implementing the code.

Status:

(THIS ANNEX IS NOT PART OF THE RECOMMENDATION)

Coefficients of $g(x)$:Polynomial in a :

	a^7	a^6	a^5	a^4	a^3	a^2	a	1
$G_0 = G_{32} = a^0$	0	0	0	0	0	0	0	1
$G_1 = G_{31} = a^{249}$	0	1	0	1	1	0	1	1
$G_2 = G_{30} = a^{59}$	0	1	1	1	1	1	1	1
$G_3 = G_{29} = a^{66}$	0	1	0	1	0	1	1	0
$G_4 = G_{28} = a^4$	0	0	0	1	0	0	0	0
$G_5 = G_{27} = a^{43}$	0	0	0	1	1	1	1	0
$G_6 = G_{26} = a^{126}$	0	0	0	0	1	1	0	1
$G_7 = G_{25} = a^{251}$	1	1	1	0	1	0	1	1
$G_8 = G_{24} = a^{97}$	0	1	1	0	0	0	0	1
$G_9 = G_{23} = a^{30}$	1	0	1	0	0	1	0	1
$G_{10} = G_{22} = a^3$	0	0	0	0	1	0	0	0
$G_{11} = G_{21} = a^{213}$	0	0	1	0	1	0	1	0
$G_{12} = G_{20} = a^{50}$	0	0	1	1	0	1	1	0
$G_{13} = G_{19} = a^{66}$	0	1	0	1	0	1	1	0
$G_{14} = G_{18} = a^{170}$	1	0	1	0	1	0	1	1
$G_{15} = G_{17} = a^5$	0	0	1	0	0	0	0	0
$G_{16} = a^{24}$	0	1	1	1	0	0	0	1

Note that $G_3 = G_{29} = G_{13} = G_{19}$.

Further information, including encoder block diagrams, is provided by Perlman and Lee in Reference [7].

ANNEX D:
GLOSSARY OF TERMS

Purpose:

This annex provides definitions for many of the technical terms used in the Recommendation to help clarify their meaning among users and reduce the possibility of misunderstanding among multinational implementers.

Status:

(THIS ANNEX IS NOT PART OF THE RECOMMENDATION)

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GLOSSARY OF TERMS

- Binary Symmetric Channel (BSC)** - A channel through which it is possible to send one binary digit per unit of time and for which there is a probability p ($0 < p < 1/2$), that the output is different from the input. This probability does not depend on whether the input is a zero or a one. Successive input digits are affected by the channel independently.
- Block Encoding** - A one-to-one transformation of sequences of length k of elements of a source alphabet to sequences of length n of elements of a code alphabet, $n > k$.
- Channel Symbol** - The unit of output of the innermost encoder.
- Codeblock** - A codeblock of an (n,k) block code is a sequence of n channel symbols which were produced as a unit by encoding a sequence of k information symbols, and will be decoded as a unit.
- Code Rate** - The average ratio of the number of binary digits at the input of an encoder to the number binary digits at its output.
- Codeword** - In a block code, one of the sequences in the range of the one-to-one transformation (see Block Encoding).
- Concatenation** - The use of two or more codes to process data sequentially with the output of one encoder used as the input of the next.
- Connection Vector** - In convolutional coding, a device used to specify one of the parity checks to be performed on the shift register in the encoder. For a binary constraint length k convolutional code, a connection vector is a k -bit binary number. A "one" in position i (counted from the left) indicates that the output of the i th stage of the shift register is to be used in computing that parity check.
- Constraint Length** - In convolutional coding, the number of consecutive input bits that are needed to determine the value of the output symbols at any time.
- Convolutional Code** - As used in this document, a code in which a number of output symbols are produced for each input information bit. Each output symbol is a linear combination of the current input bit as well as some or all of the previous $k-1$ bits where k is the constraint length of the code.
- GF(n)** - The Galois Field consisting of exactly " n " elements.

- Inner Code - In a concatenated coding system, the last encoding algorithm that is applied to the data stream. The data stream here consists of the codewords generated by the outer decoder.
- Modulating waveform - A way of representing data bits ("1" and "0") by a particular waveform.
- NRZ-L - A modulating waveform in which a data "one" is represented by one of two levels, and a data "zero" is represented by the other level.
- NRZ-M - A modulating waveform in which a data "one" is represented by a change in level and a data "zero" is represented by no change in level.
- Outer Code - In a concatenated coding system, the first encoding algorithm that is applied to the data stream.
- Reed-Solomon ("R-S") Symbol - A set of J bits that represents an element in $GF(2^J)$, the code alphabet of a J -bit Reed-Solomon code.
- Systematic Code - A code in which the input information sequence appears in unaltered form as part of the output codeword.
- Transparent Code - A code that has the property that complementing the input of the encoder or decoder results in complementing the output.
- Virtual Fill - In a systematic block code, a codeword can be divided into an information part and a parity (check) part. Suppose that the information part is N symbols long (a symbol is defined here to be an element of the code's alphabet) and that the parity part is M symbols long. A "shortened" code is created by taking only S ($S < N$) information symbols as input, appending a fixed string of length $N-S$ and then encoding in the normal way. This fixed string is called "fill." Since the fill is a predetermined sequence of symbols, it need not be transmitted over the channel. Instead, the decoder appends the same fill sequence before decoding. In this case, the fill is called "Virtual Fill." $\square$